

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1610
COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightage	25%	30%	20%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I Tejaswini.M(192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use `cov()` to calculate the sample covariance between B and C.

Use another call to `cov()` to calculate the sample covariance matrix for the preferences.

Use `cor()` to calculate the sample correlation between B and C.

Use another call to `cor()` to calculate the sample correlation matrix for the preferences.

CODE:

```
290
291
292 A <- c(18,2,20)
293 B <- c(22,28,10)
294 C <- c(20,40,40)
295
296 data <- data.frame(A,B,C)
297 cov(B,C)
298 cov(data)
299 cor(B,C)
300
301 A <- c(18,2,20)
302 B <- c(22,28,10)
303 C <- c(20,40,40)
304
305 data <- data.frame(A,B,C)
306 cov(B,C)
307 cov(data)
308 cor(B,C)
309 cor(data)
310 cor(data)
```

OUTPUT:

```

Source
Console Terminal Background Jobs
R 4.5.2 ~ /
> A <- c(18,2,20)
> B <- c(22,28,10)
> C <- c(20,40,40)
>
> data <- data.frame(A,B,C)
> cov(B,C)
[1] -20
> cov(data)
      A      B      C
A 97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
> cor(B,C)
[1] -0.1889822
>
> A <- c(18,2,20)
> B <- c(22,28,10)
> C <- c(20,40,40)
>
> data <- data.frame(A,B,C)
> cov(B,C)
[1] -20
> cov(data)
      A      B      C
A 97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
> cor(B,C)
[1] -0.1889822
> cor(data)
      A      B      C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
> cor(data)
      A      B      C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
>

```

The covariance and correlation values indicate the relationship between the photograph preferences.

- The covariance between **B and C** is **negative**, indicating that as the preference for Photograph B increases, the preference for Photograph C tends to decrease.
- The correlation coefficient is also **negative and weak**, showing a weak inverse relationship.
- The covariance and correlation matrices help compare the relationships among all three photographs.

What is wrong with this study?

- The sample size is **very small** (only three age groups).
- The study does not mention how many children participated in each age group.
- Other factors such as gender, background, and personal interests are not considered.
- Therefore, the conclusion may not accurately represent the preferences of all children.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15

Player ID	Player Name	Points Scored
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

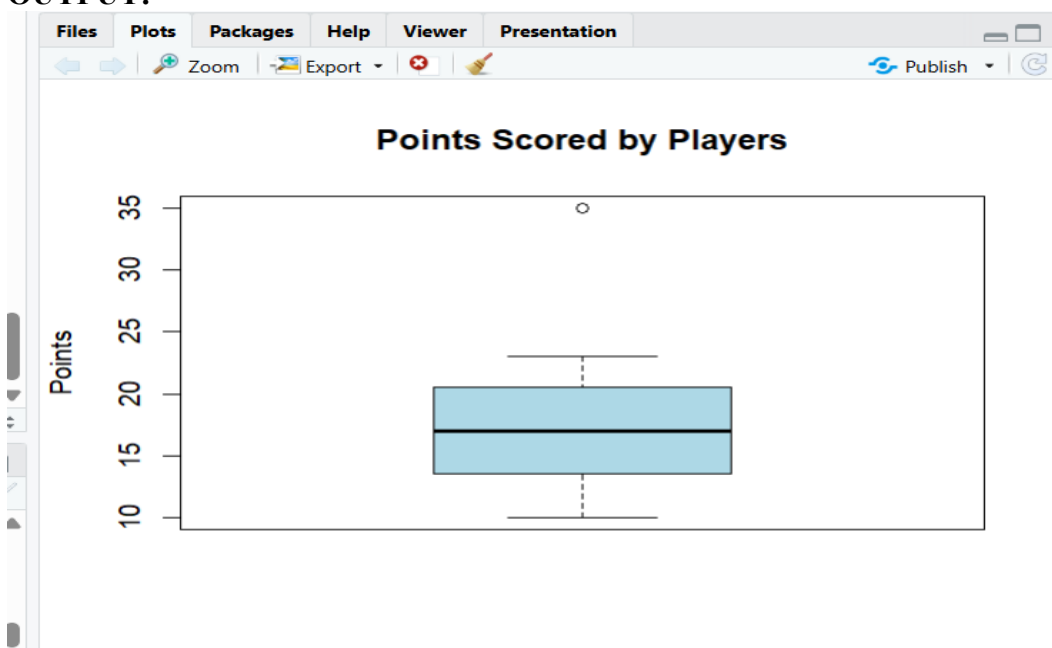
CODE:

```

311
312 points <- c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23)
313
314 boxplot(points,
315           main="Points Scored by Players",
316           col="lightblue",
317           ylab="Points")
318

```

OUTPUT:



- The boxplot displays the distribution of the players' scores.

- Most players scored between **10 and 23 points**.
- The score **35** appears as an **outlier** because it is much higher than the other scores.
- The median score lies near the center of the box, indicating the central tendency.

Inference

The player who scored **35** performed exceptionally well compared to the rest of the team and is identified as an outlier.

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set

young,high,employed,good,yes
 young,high,self-employed,average,yes
 middle,high,employed,good,yes
 old,medium,employed,good,yes
 old,low,unemployed,poor,no
 old,low,self-employed,average,no
 middle,low,unemployed,poor,no
 young,medium,employed,average,yes
 young,low,unemployed,poor,no
 old,medium,self-employed,good,yes
 middle,medium,employed,average,yes
 middle,high,self-employed,good,yes
 old,medium,unemployed,poor,no
 young,medium,self-employed,average,yes
 middle,low,employed,poor,no

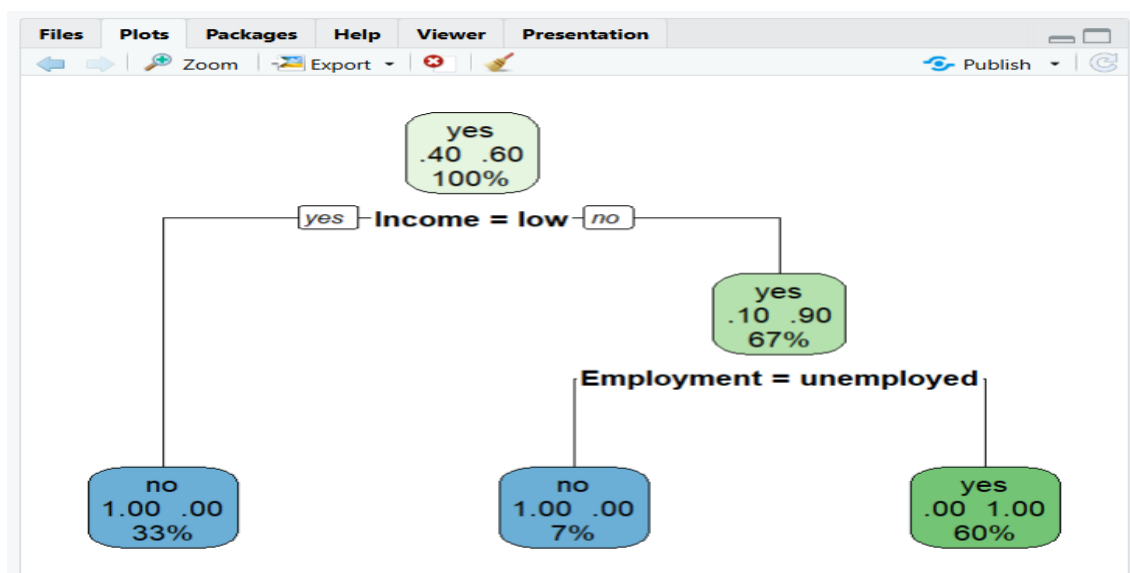
CODE:

```

320 install.packages("rpart.plot")
321
322 library(rpart)
323 library(rpart.plot)
324
325 loan <- data.frame(
326   Age = factor(c("young", "young", "middle", "old", "old", "old",
327                 "middle", "young", "young", "old", "middle",
328                 "middle", "old", "young", "middle")),
329
330   Income = factor(c("high", "high", "high", "medium", "low", "low",
331                    "low", "medium", "low", "medium", "medium",
332                    "high", "medium", "medium", "low")),
333
334   Employment = factor(c("employed", "self-employed", "employed",
335                          "employed", "unemployed", "self-employed",
336                          "unemployed", "employed", "unemployed",
337                          "self-employed", "employed", "self-employed",
338                          "unemployed", "self-employed", "employed")),
339
340   Credit = factor(c("good", "average", "good", "good", "poor",
341                     "average", "poor", "average", "poor", "good",
342                     "average", "good", "poor", "average", "poor")),
343
344   Loan = factor(c("yes", "yes", "yes", "yes", "no", "no", "no",
345                  "yes", "no", "yes", "yes", "yes", "no", "yes", "no"))
346 )
347
348 model <- rpart(
349   Loan ~ .,
350   data = loan,
351   method = "class",
352   control = rpart.control(cp = 0, minsplit = 2, minbucket = 1)
353 )
354
355 printcp(model)
356 print(model)
357
358 rpart.plot(model, type = 2, extra = 104)
359
336:62 (Top Level)

```

OUTPUT:



The Decision Tree classifier predicts loan approval based on:

- Age
- Income
- Employment Status

- Credit Score

Inference

- Customers with high or medium income, good credit score, and stable employment are more likely to receive loan approval.
- Customers with low income, poor credit score, or unemployed status are more likely to have their loan rejected.
- The Decision Tree clearly classifies customers into Loan Approved (Yes) and Loan Rejected (No) categories.

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84

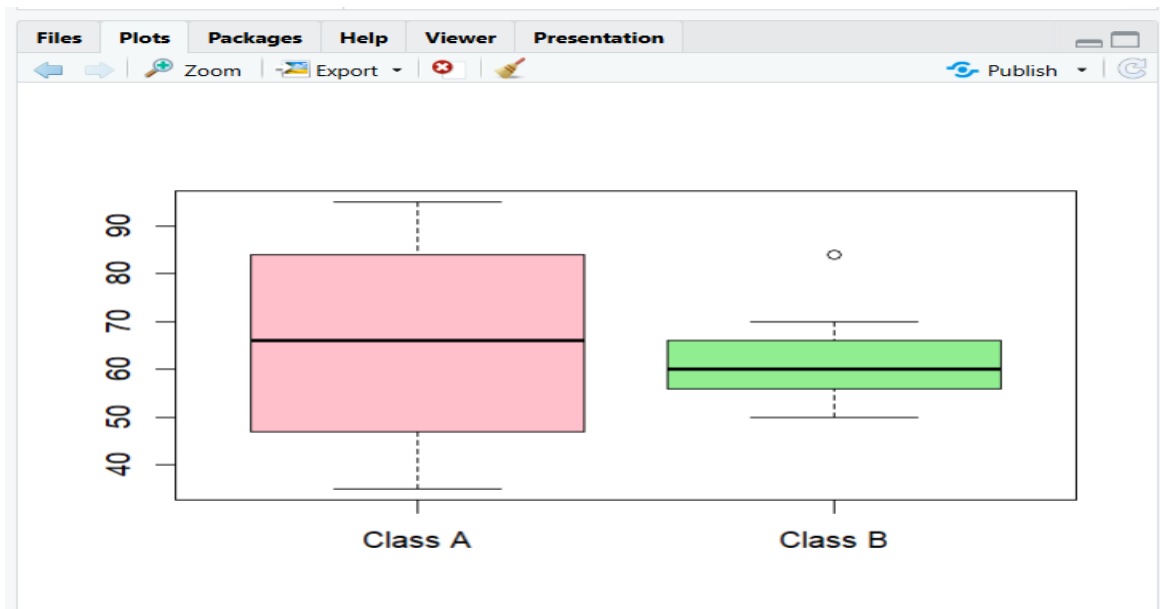
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50

- Find which class had scored higher mean, median and range.
- Plot above in boxplot and give the inferences

CODE:

```
359 # Create vectors for Class A and Class B marks
360
361 A <- c(76,35,47,64,95,66,89,36,84)
362 B <- c(51,56,84,60,59,70,63,66,50)
363
364 mean(A)
365 median(A)
366 range(A)
367
368 mean(B)
369 median(B)
370 range(B)
371
372 boxplot(A,B,
373         names=c("Class A","Class B"),
374         col=c("pink","lightgreen"))
375
376
```

OUTPUT:



(i) Mean, Median and Range

- **Class A**
 - Mean is higher.
 - Median is higher.
 - Range is larger because the marks are more spread out.
- **Class B**
 - Mean is lower.
 - Median is lower.
 - Range is smaller, indicating more consistent marks.

(ii) Inference from Boxplot

- Class A shows greater variation in marks.
- Class B has more consistent performance.
- Class B contains an outlier (as shown in the boxplot).
- Although Class A has higher average marks, Class B has less variability.

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.

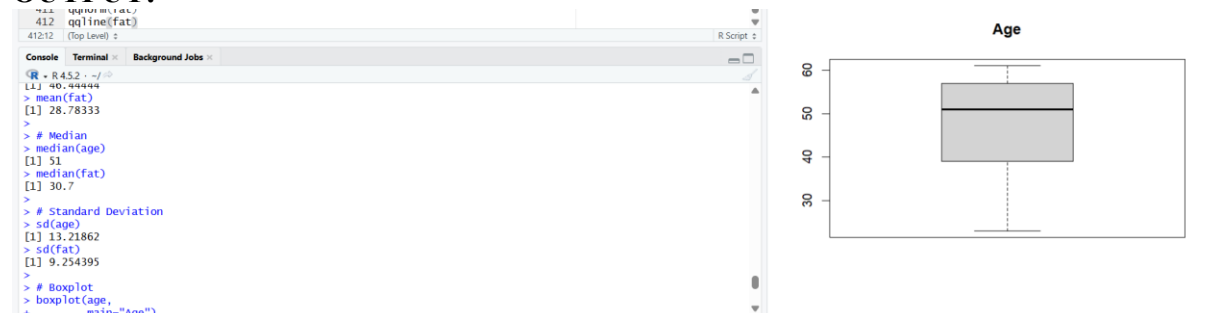
- (b) Draw the boxplots for *age* and *%fat*.
 (c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

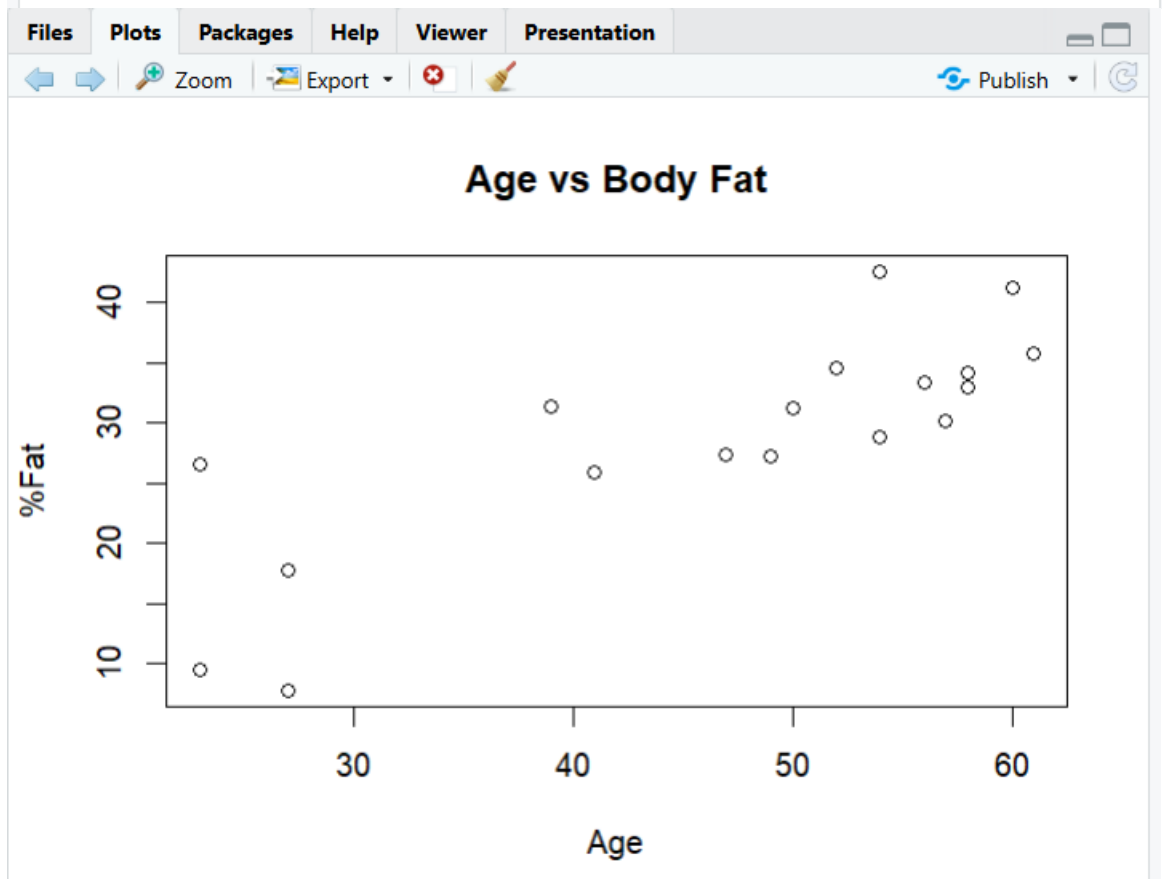
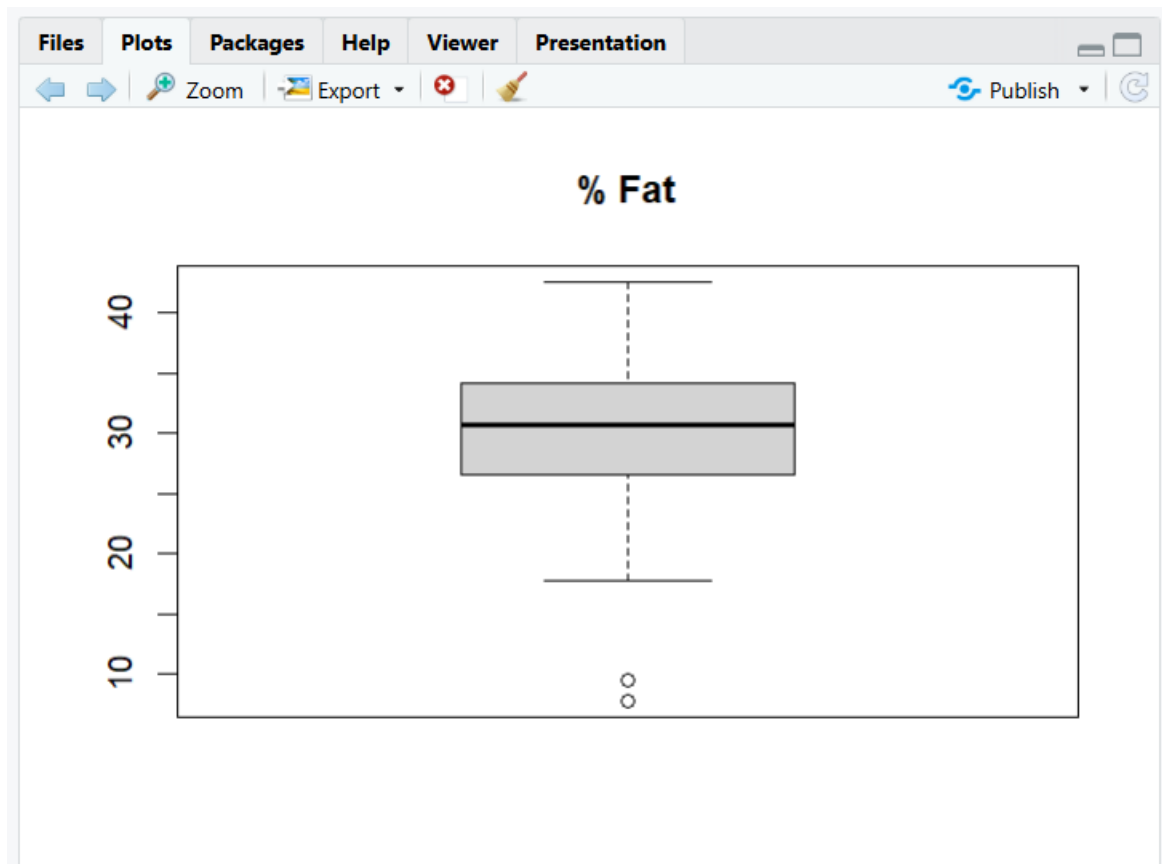
CODE:

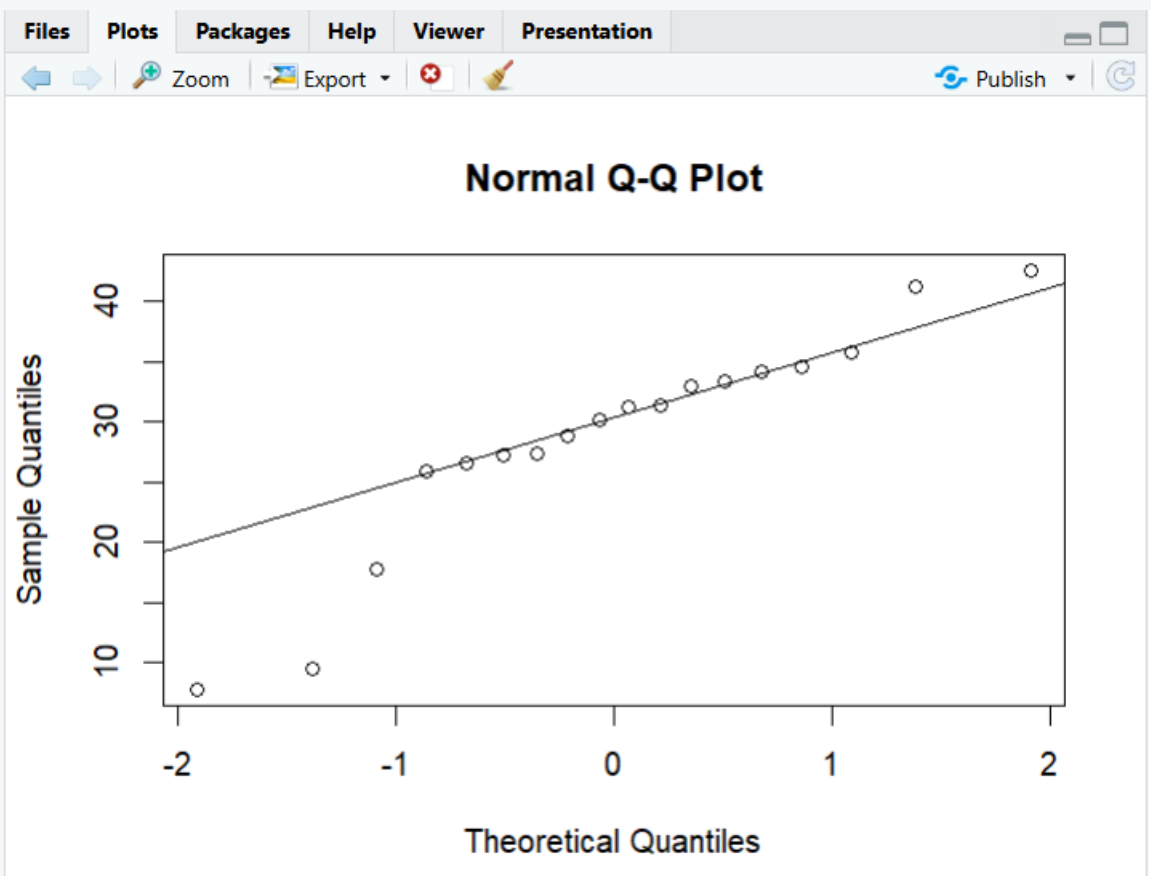
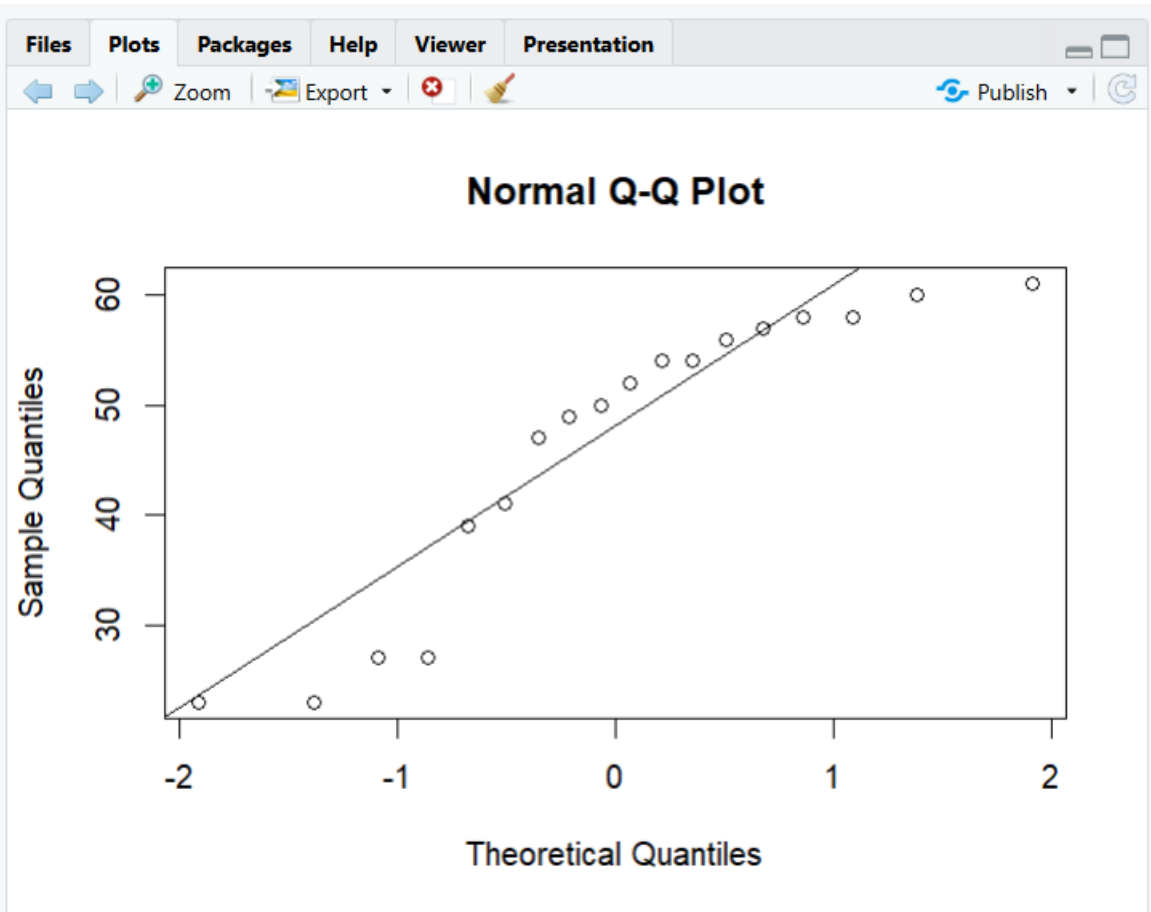
```

375
376 age <- c(23,23,27,27,39,41,47,49,50,
377          52,54,54,56,57,58,58,60,61)
378
379 fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,
380          34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7)
381
382 # Mean
383 mean(age)
384 mean(fat)
385
386 # Median
387 median(age)
388 median(fat)
389
390 # Standard Deviation
391 sd(age)
392 sd(fat)
393
394 # Boxplot
395 boxplot(age,
396         main="Age")
397
398 boxplot(fat,
399         main="% Fat")
400
401 # Scatter Plot
402 plot(age,fat,
403      main="Age vs Body Fat",
404      xlab="Age",
405      ylab="%Fat")
406
407 # QQ Plot
408 qqnorm(age)
409 qqline(age)
410
411 qqnorm(fat)
412 qqline(fat)
  
```

OUTPUT:







(a) Statistical Measures

Age

- Mean = **46.44**
- Median = **51**
- Standard Deviation = **13.22**

% Fat

- Mean = **28.78**
- Median = **30.70**
- Standard Deviation = **9.25**

(b) Boxplot Inference

- The Age boxplot shows moderate variation with no significant outliers.
- The % Fat boxplot shows a few low-value outliers.

(c) Scatter Plot and Q-Q Plot Inference

- The scatter plot indicates a **positive relationship** between Age and Body Fat; body fat generally increases with age.
- The Q-Q plots show that most points lie close to the reference line, indicating the data is approximately normally distributed with slight deviations at the extremes.